

Week - 04 - Exceptions, Dataflow\Lecture_8_code_1.py

```
1 import math
2 G = 6.674e-11
3
4 def escape(M, r, V):
5     '''
6     V is leaving object's velocity
7     M is planet's mass
8     r is planet's radius
9     calculate v
10    v is escape velocity
11    e is excess speed '''
12
13    v = (2*G*M / r)**0.5
14
15    # e = math.sqrt(-5)
16    e = (V**2 - v**2)**0.5
17
18
19    if V > v:
20        e = math.sqrt(V**2 - v**2)
21        print(f'Excess speed: {e}')
22    else:
23        print("The object won't escape")
24
25 escape(7.3e22, 1.7e6, 5000)
26 escape(6e24, 5.6e6, 5000)
```